(Figure 7.8). Within the shealth, the bacteria are capable of growth and division. Aquatic bacteria mostly form sheath. Examples of sheathed bacteria include *Leptothrix discophora* (also known as iron bacteria), *Sphaerotilus* and *Clonothrix*.

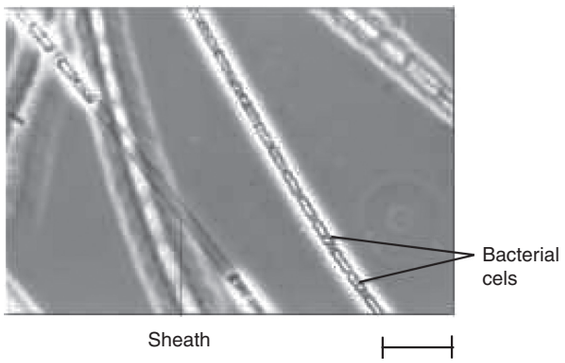


Figure 7.8: Sheathed bacterium

Function:

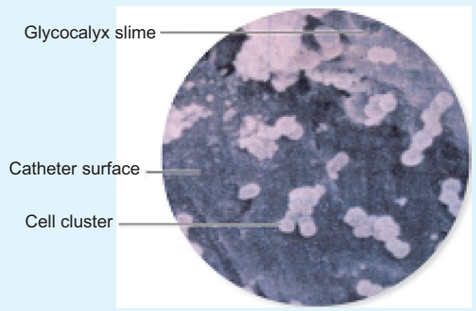
- It provides mechanical support.
- In a few bacteria, shealth is strengthened by the deposition of ferric and manganese hydroxides.

Prosthecae

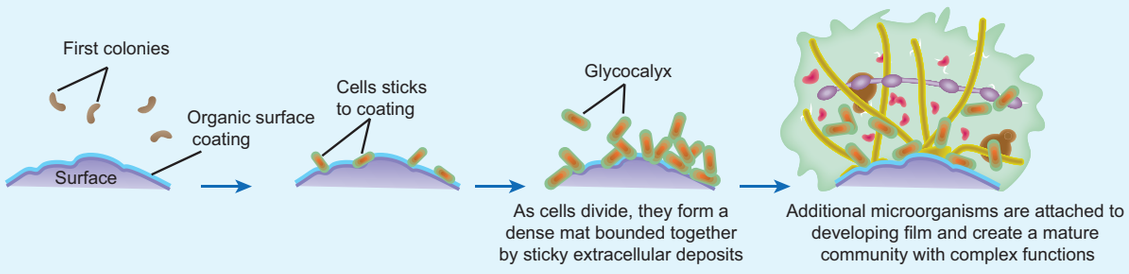
They are semi rigid extensions of cell wall and cell membrane. Some bacteria may contain more than one prosthecae (Figure 7.9). Aerobic bacteria in fresh water and marine environment possess prosthecae. Some of the prosthecate bacteria are *Caulobacter, Stellar, Prosthecobacter and Hyphomicrobium*.

Infobits

Biofilms:



Microbial adhesion to animate or inanimate surfaces can be mediated by polysaccharides capsules or slime. These adherence polymers are collectively called as adhesions. Microorganism tend to adhere to any surface and the layer they produce is called Biofilm. Biofilm can be harmful or beneficial to humans. Biofilm formation is a critical issue for almost all surfaces in health care and food preparation settings. Biofilms may form on a wide variety of surfaces, including living tissues, medical devices, industrial or portable water piping system, etc., Biofilm formation is a multi-step process starting with attachment to a surface, then formation of three dimensional structure and finally ending with maturation and detachment. During biofilm formation many species of bacteria are able to communicate with one another through specific mechanism called quorum sensing.



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